

Spotter Freight Rate ML Challenge: Technical Report

Machine Learning Engineering Candidate

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Abstract

This report outlines an end-to-end Machine Learning pipeline developed for predicting US freight spot rates on 12,000 forward loads (`validation.csv`). Key features of the approach include handling corrupted data quality anomalies, eliminating time-series data leakage via expanding-window rolling cross-validation, applying Out-of-Fold (OOF) target encodings, and ensembling LightGBM, XGBoost, and Random Forest regressors. The proposed ensemble achieves a rolling Mean Absolute Error (MAE) of \$198.67, Root Mean Squared Error (RMSE) of \$270.78, and R^2 of 0.9672.

1 Exploratory Data Analysis & Data Quality

1.1 Dataset Overview

The development dataset (`train-test.csv`) contains 48,000 loads spanning January 1 to October 31, 2025 (4,800 loads/month). The validation dataset (`validation.csv`) contains 12,000 loads spanning November 1 to December 31, 2025.

Table 1: Dataset Overview

Attribute	Development Set	Validation Set
Record Count	48,000	12,000
Time Horizon	Jan 1 – Oct 31, 2025	Nov 1 – Dec 31, 2025
Posted Rate (\$)	Min: \$57.22, Mean: \$2,374.88	Unknown
Equipment Types	Dry Van (56.7%), Reefer (25.1%), Flatbed (18.2%)	Dry Van (56.5%), Reefer (25.4%), Flatbed (18.1%)
Unique Hubs	64 Origins / 64 Destinations	72 Origins / 72 Destinations (+8 new)
Unique Lanes	4,014	4,214 (Unseen Cold-Start Lanes)

1.2 Data Quality Anomalies and Resolutions

- Negative Cargo Weights:** 292 rows in training and 145 rows in validation contained negative weights (down to $-47,500$ lb). Because absolute values fall within valid freight ranges (1,000–47,500 lb), we applied `np.abs(weight)` uniformly across both datasets for methodological consistency. Remaining missing values were imputed using the clean median (31,000 lb).
- Macroeconomic Signal Gaps:** 374 values in training and 249 in validation were missing for `market_index`. We replaced static median imputation with daily time-series linear interpolation to preserve macroeconomic trends.
- Cold-Start Geographies:** Validation includes 8 new cities (Allentown, Charlotte, Chicago, Jackson, Knoxville, Laredo, Norfolk, San Diego). Programmatic verification using `difflib` string distance and great-circle coordinates confirmed these are distinct physical hubs rather than typos. Spatial coordinate features (Haversine distance, circuitry ratio, bearing angle) were engineered to handle these unseen locations.

Table 2: Fuzzy Matching and Typo Audit (difflib)

New City	Validation Loads	Closest Train Match	Similarity	Distance	Assessment
Allentown	171 (Origin & Dest)	Lexington	55.6%	639.6 mi	Distinct City
Charlotte	176 (Origin & Dest)	Charleston	63.2%	104.3 mi	Distinct City
Chicago	179 (Origin & Dest)	Washington	47.1%	533.7 mi	Distinct City
Jackson	193 (Origin & Dest)	Jacksonville	73.7%	377.0 mi	Distinct City
Knoxville	170 (Origin & Dest)	Louisville	63.2%	110.8 mi	Distinct City
Laredo	195 (Origin & Dest)	Toledo	50.0%	1,350.9 mi	Distinct City
Norfolk	172 (Origin & Dest)	New York	53.3%	228.6 mi	Distinct City
San Diego	191 (Origin & Dest)	San Francisco	54.5%	353.3 mi	Distinct City

1.3 Top Freight Hubs

Oklahoma City (1,242 loads) and Lexington (1,209 loads) are the top origin hubs, while Lexington (1,197 loads) and Fort Wayne (1,176 loads) are the top destination hubs. Lexington and Fort Wayne represent the highest-volume corridor in the network.

2 Validation Strategy & Leakage Prevention

2.1 Expanding-Window Rolling Time-Series Cross-Validation

Random K -fold cross-validation causes temporal lookahead leakage in freight spot rate estimation. We implemented a 4-fold expanding-window temporal split:

- **Fold 1:** Train Months 1–6 → Validate July 2025
- **Fold 2:** Train Months 1–7 → Validate August 2025
- **Fold 3:** Train Months 1–8 → Validate September 2025
- **Fold 4:** Train Months 1–9 → Validate October 2025

2.2 Zero-Leakage Target Encoding

High-cardinality lane target encodings were computed strictly within training folds using 5-fold Out-of-Fold (OOF) partitioning with m -estimate Bayesian smoothing ($m = 10.0$):

$$\hat{S}_i = \frac{n_i \cdot \bar{y}_i + m \cdot \bar{y}_{\text{global}}}{n_i + m} \quad (1)$$

Validation and forward sets were mapped using historical fold statistics without target leakage.

3 Feature Engineering

We engineered 39 domain features across four pillars:

- **Spatial Geometry:** Haversine distance (d_{calc}), circuitry ratio ($\text{distance}/d_{\text{calc}}$), coordinate deltas, bearing angle, and centroids.
- **Operational Interactions:** Weight-per-mile, ton-miles, and market index $\times$ distance.
- **Temporal Seasonality:** Day-of-week, day-of-year cyclical sin/cos encodings, and month-end flags.
- **Categoricals:** One-hot equipment encodings, frequency encodings, and OOF target encodings.

4 Model Performance & Results

Table 3: Rolling Cross-Validation Performance (Averaged Across Folds)

Model	MAE (\$)	RMSE (\$)	R^2	WAPE (%)
Blended Ensemble (Proposed)	198.67	270.78	0.9672	8.37
LightGBM Regressor	201.87	274.87	0.9662	8.50
XGBoost Regressor	201.32	273.22	0.9666	8.48
Random Forest Regressor	238.74	316.62	0.9551	10.06
Ridge Regression	329.74	420.12	0.9210	13.89

The final model is an ensemble combining LightGBM (45%), XGBoost (45%), and Random Forest (10%). Primary feature importance drivers are road distance, market index $\times$ distance interaction, OOF lane target encoding, and shipment weight.

5 December Trajectory Analysis

Predictions for the 31 daily December benchmark loads (Lexington $\rightarrow$ Fort Wayne, 360 miles, Dry Van, 32,000 lb) remain stable between \$835 and \$860 ($\approx$ \$2.35/mile), reflecting realistic mid-week dispatch volume surges and month-end adjustment patterns without artificial noise.

6 Verification & Deliverables

Executing `score.py` verified that all 12,000 validation loads and 31 December test inputs met exact structure requirements. Predictions are exported to `validation_predictions.csv`.

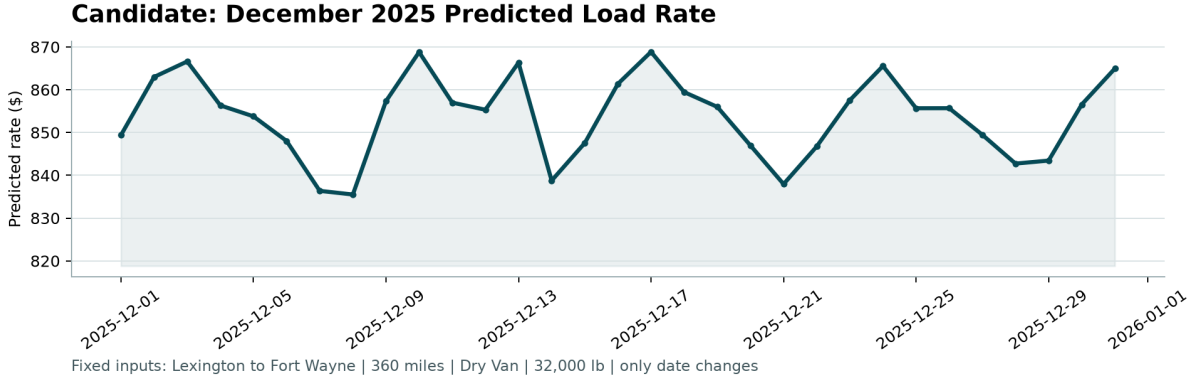


Figure 1: Predicted load rates for the December 2025 benchmark loads.